

Word Counter Agent

Build and deploy your first Apify Actor
AI Agents Classroom Series • thecoachdinesh.com

What does this Actor do?

This Apify Actor is a cloud-based script that reads any web page and counts how many times a specific word appears on it.

Input: A URL and a word to search for

Output: The count, the page title, and 5 sample sentences where the word appears

Example: Search for the word **AI** on thecoachdinesh.com/courses/ai/reference.html — the Actor returns **312 occurrences**



What you need before starting:

- A laptop (Mac or Windows)
- A Google account (Gmail)
- Internet connection
- 30 minutes

PART 1 — Register on Apify (5 minutes)

Apify is the platform where we will build and run our actor. Registration is free — no credit card needed. You get \$5 of free compute credits per month, more than enough for this exercise.



Step 1 — Scan the QR code with your phone camera

Or open your browser and go to:

<https://apify.com/sign-up>

Step 2 — Click **Continue with Google**

Step 3 — Select your Gmail account

Step 4 — You will land on the Apify Console dashboard

■ **Free tier:** Apify free tier gives \$5 compute credits per month. No credit card needed. This exercise uses less than \$0.01 of credits.

PART 2 — Create a New Actor (3 minutes)

Once you are in the Apify Console, create a new actor by following these clicks:

1	Click Actors in the left sidebar You will see a list of actors (empty for new accounts)
2	Click Create new button Top right corner of the Actors page
3	What do you want to build? → Click Web scraper We are reading a web page to extract data
4	Which language? → Click JavaScript Our code is written in JavaScript
5	Best match → Click JavaScript one-page scraper We only scrape one URL at a time
6	Where to host? → Click Host on Apify Everything stays in the browser — no Terminal needed
7	Name your actor Type: word-counter in the name field at the top

■ **You will land on the Web IDE:** A code editor inside the browser with a file panel on the left. This is where you will paste all the code.

PART 3 — Add the Code (15 minutes)

You need to update 4 files. The file panel is on the LEFT side of the Web IDE.

For each file: click it → Cmd+A to select all → Delete → paste new code → Save.

File	Where in the panel	What it does
src/main.js	src folder → main.js	The main actor code — reads the page and counts words
.actor/input_schema.json	.actor folder → input_schema.json	Defines the input form (URL, word, case sensitive)
package.json	Root level (not in any folder)	Lists the libraries the code needs to run
Dockerfile	Root level (not in any folder)	Tells Apify how to build and run the actor
output_schema.json	DO NOT CHANGE	Leave the default — our actor uses Apify default dataset

■■ **Hidden folder tip:** The .actor folder starts with a dot so Mac hides it. In Finder press Cmd+Shift+. (dot key) to show hidden files and folders.

File 1 of 4 — src/main.js (The main actor code)

1	Click src folder in the left panel Expand it by clicking the triangle
2	Click main.js The file opens in the code editor
3	Select all existing code Press Cmd+A (Mac) or Ctrl+A (Windows)
4	Delete it Press Delete or Backspace — the editor should be empty
5	Copy the code below Select all code in the green block below and copy it
6	Paste Click inside the editor and press Cmd+V or Ctrl+V
7	Save Click the Save button in the top right of the editor

Paste ALL of this into src/main.js:

```
// =====  
// Word Counter Actor – thecoachdinesh.com  
// Reads a URL, counts occurrences of a given word on the page  
// =====  
  
// INPUT:  
  
// url (string) – the page to scan (default: thecoachdinesh reference page)  
// word (string) – the word to search for (case-insensitive)  
//  
  
// OUTPUT (dataset):  
// { url, word, count, caseSensitiveCount, positions, pageTitle }  
// =====  
  
import { Actor } from 'apify';  
import { CheerioCrawler } from 'crawlee';  
  
await Actor.init();  
  
// ■■ 1. Read input ████████████████████████████████████████████████████████  
  
const input = await Actor.getInput();  
  
const {  
url = 'https://www.thecoachdinesh.com/courses/ai/reference.html',  
word = 'AI',  
caseSensitive = false,
```

```

} = input || {};

if (!word || word.trim() === '') {
  throw new Error('Input "word" is required. Please provide a word to search for.');
```

```
snippets.push('... + snippetMatch[0].trim() + '...');
snippetCount++;
}

result = {
url: request.url,
pageTitle,
wordSearched: searchWord,
caseSensitive,
count,
caseSensitiveCount,
note: caseSensitive
? 'Count is case-sensitive'
: 'Count is case-insensitive (e.g. "AI", "Ai", "ai" all counted)',
sampleContexts: snippets,
};

console.log(`■ Found "${searchWord}" ${count} times on the page`);
console.log(` Page title: "${pageTitle}"`);
},

failedRequestHandler({ request, error }) {
console.error(`■ Failed to load: ${request.url} — ${error.message}`);
},
});

await crawler.run([url]);

// ■■ 3. Push result to dataset ████████████████████████████████████████████
if (result) {
await Actor.pushData(result);
console.log('\n■ RESULT:');
console.log(` Word: "${result.wordSearched}"`);
console.log(` Count: ${result.count} (case-insensitive)`);
console.log(` Count: ${result.caseSensitiveCount} (case-sensitive)`);
} else {
await Actor.pushData({
url,
wordSearched: searchWord,
count: 0,
error: 'Page could not be loaded or no text found.',
});
}

await Actor.exit();
```

Code Explanation — src/main.js line by line

You do not need to memorise this. Read through it once to understand what each part does.

Code / Line	What it does
<code>import { Actor } from 'apify'</code>	Loads the Apify SDK — gives us <code>Actor.init()</code> , <code>Actor.getInput()</code> and <code>Actor.pushData()</code>
<code>import { CheerioCrawler } from 'crawlee'</code>	Loads the web crawling library — this actually fetches and parses the HTML page
<code>await Actor.init()</code>	Initialises the actor. Required at the start of every Apify actor
<code>const input = await Actor.getInput()</code>	Reads the input the user provided — url, word, caseSensitive
<code>const { url, word, caseSensitive } = input</code>	Destructures input into variables. Sets defaults if user did not fill in a field
<code>new CheerioCrawler({ maxRequestsPerCrawl: 1 })</code>	Creates a fast HTML crawler. <code>maxRequestsPerCrawl:1</code> means only load one page
<code>\$('script, style, nav, footer').remove()</code>	Removes non-content elements so we only count words in actual readable text
<code>const text = \$('body').text()</code>	Extracts all visible text from the page body as a single string
<code>new RegExp(`\b\${escaped}\b`, flags)</code>	Creates a regex with word boundaries — AI does not match AIED or AI-powered
<code>text.match(regex) []</code>	Finds all regex matches in the text — returns an array of found words
<code>snippets — context windows</code>	Collects up to 5 sentences showing the word with 60 characters of context on each side
<code>await Actor.pushData(result)</code>	Saves the result to Apify dataset — this is the final output the user sees
<code>await Actor.exit()</code>	Cleanly shuts down the actor when finished

File 2 of 4 — .actor/input_schema.json

Hidden folder: The .actor folder is hidden on Mac. Press Cmd+Shift+. in Finder to make it visible. In the Apify Web IDE it is visible in the left panel.

- 1 Click the .actor folder**
In the left panel — expand it
- 2 Click input_schema.json**
The file opens in the editor
- 3 Select all and delete**
Cmd+A then Delete
- 4 Paste the code below**
Copy and paste the JSON below
- 5 Save**
Click Save

Paste ALL of this into .actor/input_schema.json:

```
{
  "title": "Word Counter Input",
  "type": "object",
  "schemaVersion": 1,
  "properties": {
    "url": {
      "title": "Page URL",
      "type": "string",
      "description": "The URL of the web page to scan for word occurrences.",
      "default": "https://www.thecoachdinesh.com/courses/ai/reference.html",
      "prefill": "https://www.thecoachdinesh.com/courses/ai/reference.html",
      "editor": "textfield"
    },
    "word": {
      "title": "Word to search",
      "type": "string",
      "description": "The word you want to count on the page. Case-insensitive by default.",
      "prefill": "AI",
      "editor": "textfield"
    },
    "caseSensitive": {
      "title": "Case sensitive",
      "type": "boolean",
```

```
"description": "If true, only exact case matches are counted. Default is false  
(case-insensitive).",  
"default": false,  
"editor": "checkbox"  
}  
,  
"required": ["word"]  
}
```

What this file defines:

- url — the web page to scan. Pre-filled with the thecoachdinesh reference page
- word — the word to count. Pre-filled with AI
- caseSensitive — a checkbox. Default is false (case-insensitive counting)

File 3 of 4 — package.json

1	Click package.json Root level in left panel — NOT inside any folder
2	Select all and delete Cmd+A then Delete
3	Paste the code below Copy and paste the JSON below
4	Save Click Save

Paste ALL of this into package.json:

```
{
  "name": "word-counter-actor",
  "version": "1.0.0",
  "description": "Counts occurrences of a word on any web page",
  "type": "module",
  "main": "src/main.js",
  "scripts": {
    "start": "node src/main.js"
  },
  "dependencies": {
    "apify": "^3.0.0",
    "crawlee": "^3.0.0"
  },
  "engines": {
    "node": ">=18.0.0"
  }
}
```

What this file does:

- "apify": ^3.0.0 — the Apify SDK for building and running actors
- "crawlee": ^3.0.0 — the web crawling library that provides CheerioCrawler
- "type": "module" — tells Node.js we use modern ES module import syntax

File 4 of 4 — Dockerfile

- | | |
|---|---|
| 1 | Click Dockerfile
Root level in left panel — NOT inside any folder |
| 2 | Select all and delete
Cmd+A then Delete |
| 3 | Paste the code below
Copy and paste below |
| 4 | Save
Click Save |

Paste ALL of this into Dockerfile:

```
# Apify base image with Node.js 18 + Cheerio
FROM apify/actor-node:18

# Copy package files
COPY package*.json ./

# Install dependencies
RUN npm install --omit=dev --omit=optional

# Copy source code
COPY . ./

# Run the actor
CMD npm start
```

What each line does:

- FROM apify/actor-node:18 — use Apify pre-built Node.js 18 as the base image
- COPY package*.json ./ — copy package.json first (for efficient caching)
- RUN npm install — install all libraries listed in package.json
- COPY . ./ — copy all actor code files into the container
- CMD npm start — run the actor when the container starts

output_schema.json — DO NOT CHANGE

Leave this file exactly as it is. Our actor uses Actor.pushData() which automatically saves results to the Apify default dataset. The existing output_schema.json already points there correctly.

PART 4 — Build the Actor (1 minute)

All 4 files are updated. Now build the actor. Building compiles your code and installs all dependencies on Apify servers.

1	Click the Build button Green Build button at the BOTTOM LEFT of the Web IDE
2	Wait 30 to 60 seconds You will see build logs scrolling in the Builds tab
3	Look for success You will see "Build finished" or a green checkmark
4	If build fails Re-paste each file carefully — most failures are copy-paste errors

Build failed? Go back to each file. Select all. Delete. Paste again. Make sure you copied the ENTIRE code block — scroll to the bottom to check nothing is missing.

PART 5 — Run the Actor and See Results (2 minutes)

Build succeeded? Now run the actor!

Step 1 — Click the Input tab

At the top of the editor click the Input tab. You will see a form with three fields.

Step 2 — Fill in the input form

URL	Page URL field <code>https://www.thecoachdinesh.com/courses/ai/reference.html</code>
Word	Word to search field AI (or any word you want to count)
Case	Case sensitive checkbox Leave it UNCHECKED (false)

Step 3 — Click Start

Click the orange Start button. The actor begins running immediately.

Step 4 — Watch the live logs

Click the Last run tab to see live output:

```
Searching for "AI" on: https://www.thecoachdinesh.com/courses/ai/reference.html
Case sensitive: false
Found "AI" 312 times on the page
Page title: "Reference | AI for Teachers | The Coach Dinesh"
```

Step 5 — View the results

1	Click Last run tab At the top of the actor page
2	Click Dataset In the run details panel — right side
3	Click Preview To see the output data

You should see this output:

```
{
  "url": "https://www.thecoachdinesh.com/courses/ai/reference.html",
  "pageTitle": "Reference | AI for Teachers | The Coach Dinesh",
```

```
"wordSearched": "AI",
"caseSensitive": false,
"count": 312,
"caseSensitiveCount": 287,
"note": "Case-insensitive: AI, Ai, ai all counted",
"sampleContexts": [
  "...How artificial intelligence (AI) can change the way you teach...",
  "...Today's AI applications are built for one specific task...",
  "...AI has a long way to go before supplanting a human teacher..."
]
}
```

PART 6 — Try Different Words

Go back to the Input tab, change the word, and click Start again. No need to rebuild — just re-run!

Word to search	Expected result	What this teaches you
AI	High count (~312)	Domain-specific words appear very frequently in specialist texts
teacher	Medium count	The target audience word repeats across all chapters
machine	Medium count	Technical terms appear in multiple sections
learning	Very high count	Core concept words appear in many different contexts
Google	Low count	Proper nouns and brand names appear rarely
ChatGPT	Very low count	Recent tools are mentioned briefly, not throughout
xyz123	Zero — 0	Actor correctly returns 0 when a word is not found
AI (case sensitive ONLY)	Lower than case-insensitive	ai and Ai are excluded when case-sensitive is checked

Challenge: Try searching on a DIFFERENT URL. Change the url field to any website you like and count how many times a word appears there!

Troubleshooting — Common Issues

Problem	Likely cause	Fix
Build failed	Copy-paste error in a file	Re-paste each file. Select all, delete, paste again. Check nothing is m
Cannot see .actor folder	Hidden folder on Mac	In Finder press Cmd+Shift+. to show hidden files. In the Apify Web ID
Input form shows only one field	input_schema.json not saved	Re-paste and save input_schema.json then click Build again
Count is 0 for a common word	Wrong URL or word misspelled	Try the word the — should return very high count. Check the URL is c
Actor stuck running	Network issue or wrong URL	Click Stop. Check the URL loads in your browser. Click Start again
Cannot find Dataset tab	Run has not finished yet	Wait for the run status to show SUCCEEDED then click Dataset
Build succeeded but Start is greyed out	Need to click Save after build	Make sure all files are saved then click Build again

Congratulations — you just built and deployed your first Apify Actor! ■

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